

IKEA CUSTOMER RETENTION ANALYTICS PROJECT

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1. PROJECT OVERVIEW

IKEA is a global leader in low-cost, well-designed home furnishing products. With increasing competition from online furniture brands, local retailers, and marketplaces, customer retention has become very important for long-term growth.

IKEA collects data on customer purchases, store visits, loyalty points, churn and store details. In this project I used these datasets to build an interactive Power BI dashboard to understand churn, repeat behaviour, promotions, store performance and customer lifetime value (CLV).

This project uses five main datasets:

- Customer Demographics
- Customer Transactions
- Churn Labelled Customers
- Loyalty Program
- Store Locations

2. PROBLEM STATEMENT

IKEA wants to answer some important questions:

- Why are customers becoming inactive or churning?
- Which customers are loyal and which are at risk of churn?
- Are loyalty benefits and promotions working effectively?
- How do store formats and regions affect retention?
- Which customers bring the highest Customer Lifetime Value (CLV)?

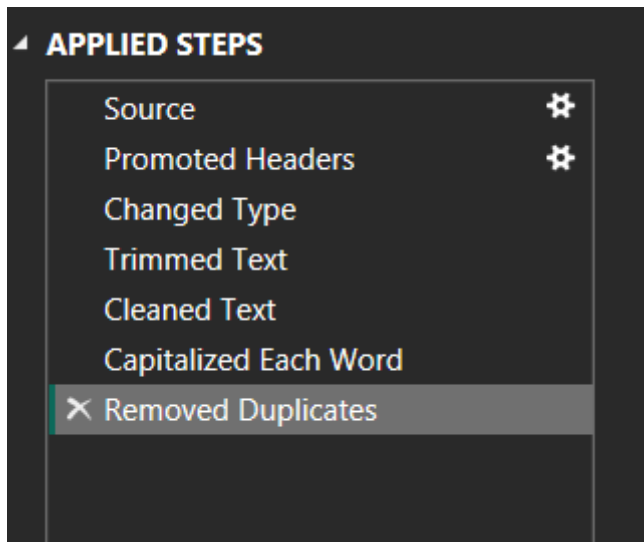
The objective of this project is to build a Power BI dashboard that helps IKEA reduce churn, improve loyalty, increase CLV and make better decisions about stores and regions.

3. TASK 1 – DATA CLEANING & DATA MODELLING

In Task 1 I cleaned all the datasets in Power Query and then created the data model in Power BI.

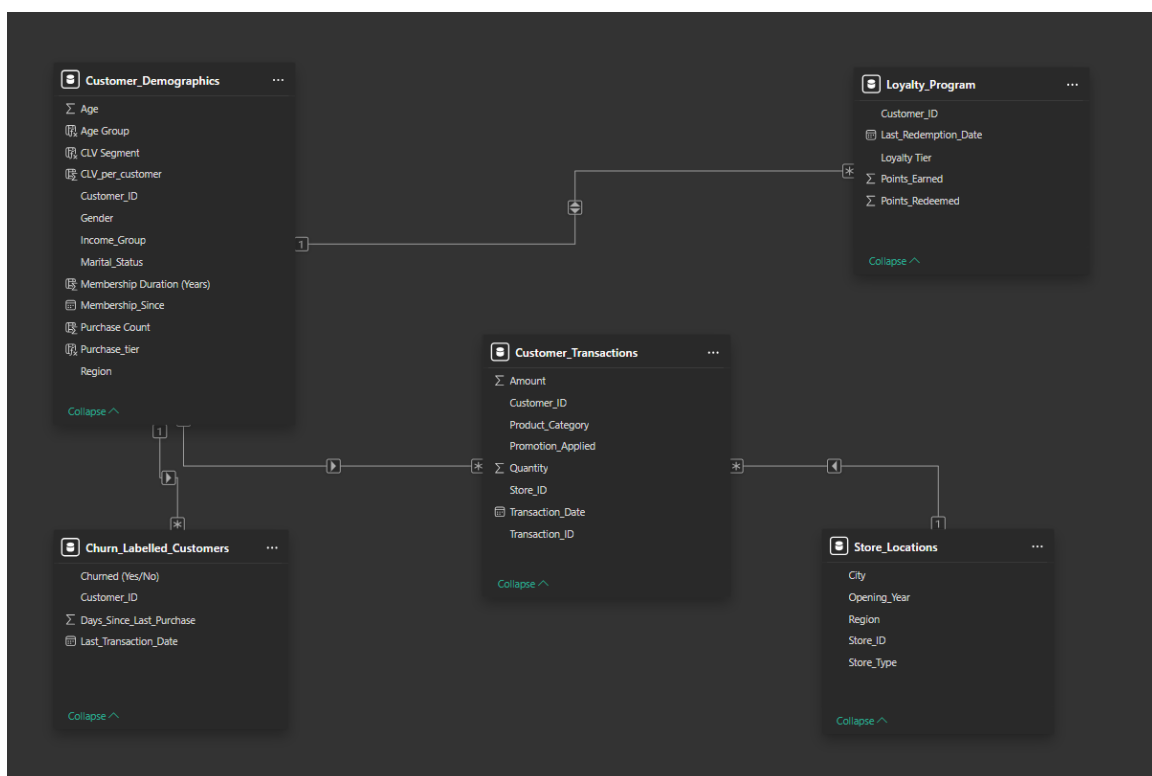
Key cleaning steps performed:

- Removed duplicate rows
- Fixed data types for date, numeric and text columns
- Trimmed and cleaned text fields
- Standardised column names
- Checked and handled null values where needed
- Verified unique keys for Customer_ID and Store_ID

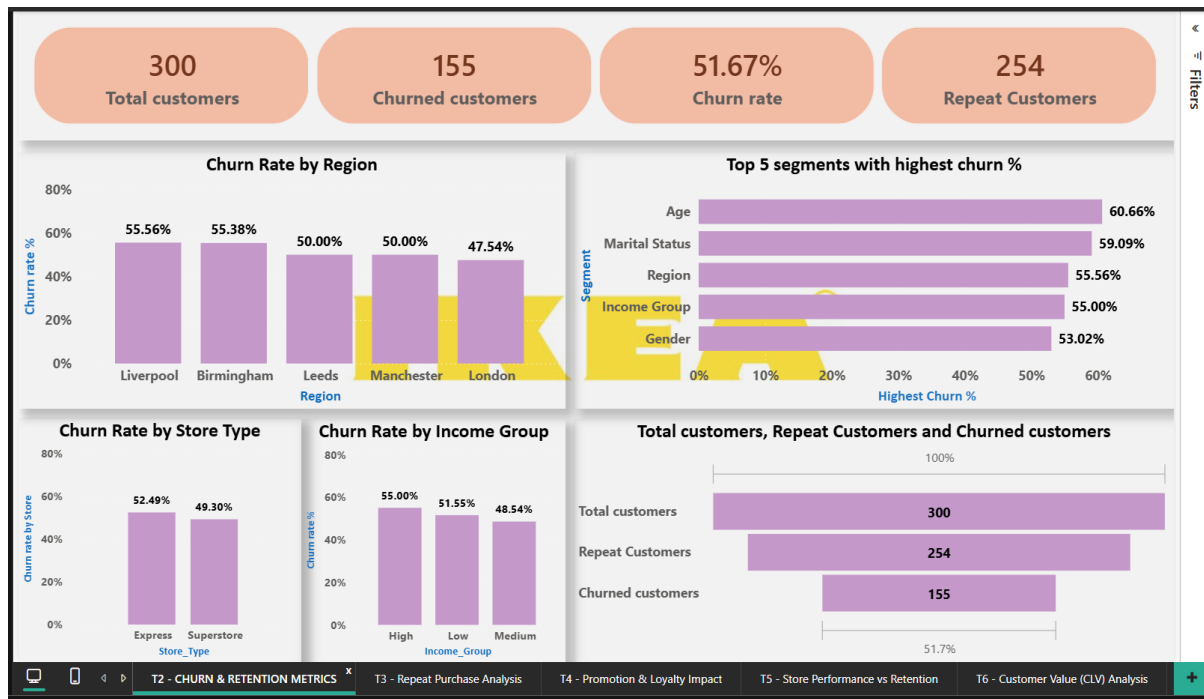


Data model relationships:

- Customer_Demographics → Customer_Transactions (1:* on Customer_ID)
- Customer_Demographics → Churn_Labelled_Customers (1:* on Customer_ID)
- Customer_Demographics → Loyalty_Program (1:* on Customer_ID)
- Store_Locations → Customer_Transactions (1:* on Store_ID)



4. TASK 2 – CHURN ANALYSIS



Key findings from churn analysis:

- Total customers = 300.
- Churned customers = 155.
- Overall churn rate ≈ 51.67%.
- Repeat customers = 254.
- Liverpool and Birmingham have the highest churn (around 55%).
- London has lower churn (~47.5%).

Business insight:

Churn is very high. IKEA should focus on reducing churn, especially in Liverpool and Birmingham.

Key DAX measures used in Task 2:

Total Customers = COUNTROWS (Customer_Demographics)

Explanation: Counts how many customer records are present in the customer table.

Churned Customers = CALCULATE(
COUNTROWS (Churn_Labelled_Customers),
Churn_Labelled_Customers[Churned (Yes/No)] = "Yes")

Explanation: Counts the number of customers whose churn flag is Yes.

Churn Rate % = DIVIDE ([Churned Customers], [Total Customers], 0)

Explanation: Calculates churn percentage as churned divided by total customers.

Repeat Customers =VAR Cust =

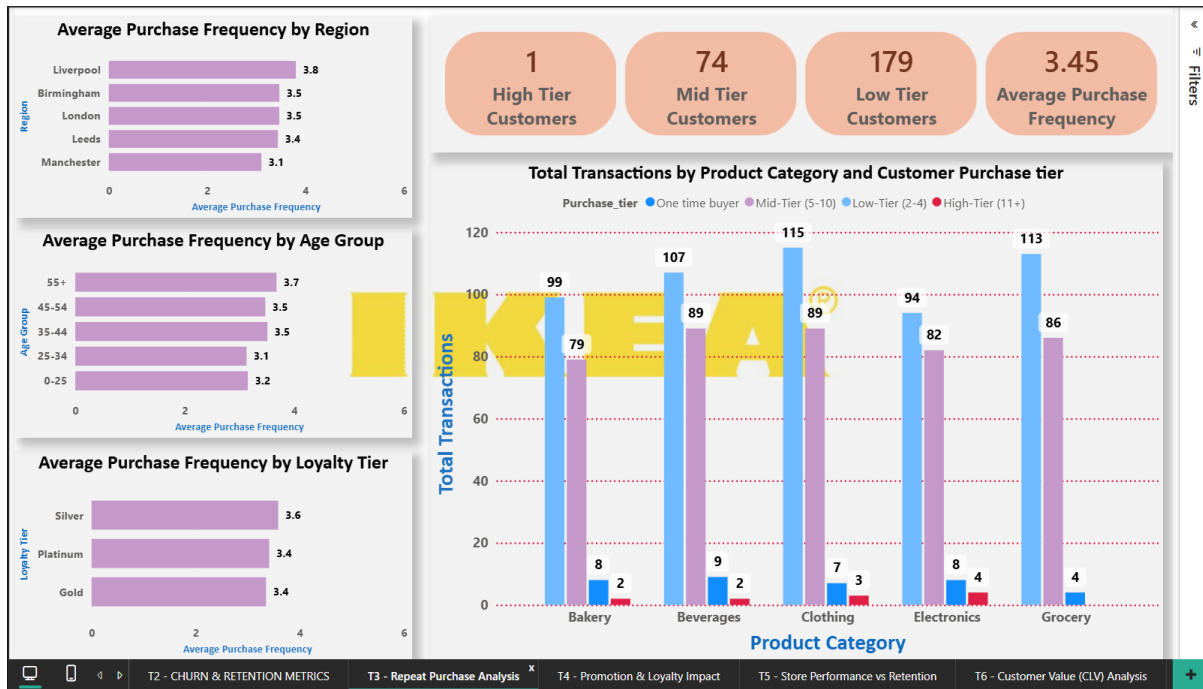
```

FILTER (
    VALUES ( Customer_Transactions[Customer_ID] ),
    CALCULATE ( COUNTROWS ( Customer_Transactions ) ) > 1
)
RETURN
COUNTROWS ( Cust )

```

Explanation: Counts customers who have made more than one transaction.

5. TASK 3 – REPEAT PURCHASE ANALYSIS



Key findings from repeat purchase analysis:

- Average purchase frequency ≈ 3.45 .
- High Tier customers = 1, Mid Tier = 74, Low Tier = 179.
- Age group 55+ has the highest average purchase frequency.
- Liverpool shows highest purchase frequency among regions.
- Silver customers buy slightly more often than Gold and Platinum.

Business insight:

Very few customers are in the High Tier. Older customers and Silver tier customers are good targets for loyalty growth and offers.

Key DAX and calculated columns used in Task 3:

Purchase Count (calculated column) =

```
COUNTROWS ( RELATEDTABLE ( Customer_Transactions ) )
```

Explanation: Counts the number of transactions for each customer.

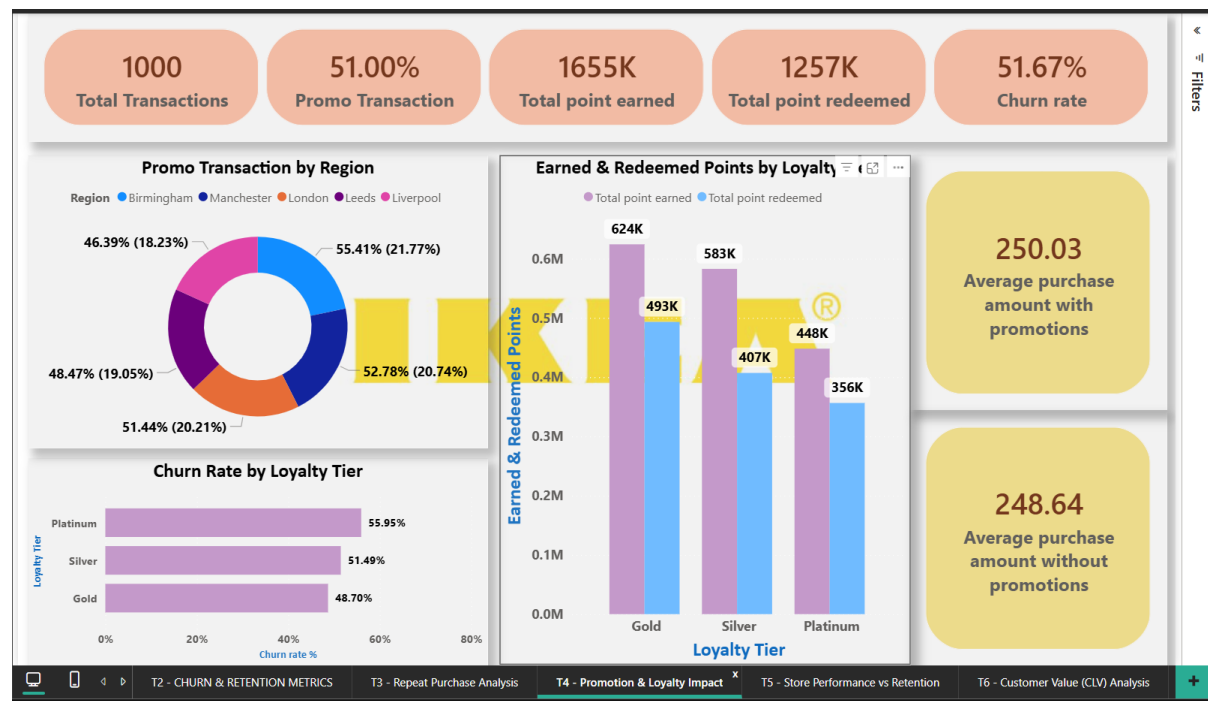
```
Purchase_tier (calculated column) = SWITCH(
    TRUE(),
    [Purchase Count] >= 11, "High-Tier (11+)",
    [Purchase Count] >= 5, "Mid-Tier (5-10)",
    [Purchase Count] >= 2, "Low-Tier (2-4)",
    [Purchase Count] = 1, "One time buyer",
    "Not a buyer"
)
```

Explanation: Classifies customers into One-time, Low, Mid and High tier based on number of purchases.

Average_Purchase_Frequency = AVERAGE (Customer_Demographics[Purchase Count])

Explanation: Calculates average number of purchases per customer in the current filter context.

6. TASK 4 – PROMOTION & LOYALTY IMPACT



Key findings from promotion and loyalty analysis:

- 51% of transactions used a promotion.
- Average amount with promotion $\approx$ 250.03.
- Average amount without promotion $\approx$ 248.64.
- Total loyalty points earned $\approx$ 2M and redeemed $\approx$ 1M.
- Gold and Silver tiers redeem the most points.
- Platinum tier has the highest churn ($\sim$ 55.95%).

Business insight:

Promotions do not reduce average spend, so they are safe to use but less effective. Loyalty points are used well, but churn is still high in Platinum and some groups, so engagement must improve.

Key DAX measures used in Task 4:

Promo Transactions =CALCULATE (
COUNTROWS (Customer_Transactions),
Customer_Transactions[Promotion_Applied] = "Yes")

Explanation: Counts how many transactions had a promotion applied.

Promo Transactions % = DIVIDE ([Promo Transactions], [Total Transactions], 0)

Explanation: Calculates the percentage of transactions where a promotion was used.

Avg Amount With Promo = CALCULATE (
AVERAGE (Customer_Transactions[Amount]),
Customer_Transactions[Promotion_Applied] = "Yes")

Explanation: Average bill amount only for transactions with promotions.

Avg Amount Without Promo = CALCULATE (
AVERAGE (Customer_Transactions[Amount]),
Customer_Transactions[Promotion_Applied] = "No")

Explanation: Average bill amount only for transactions without promotions.

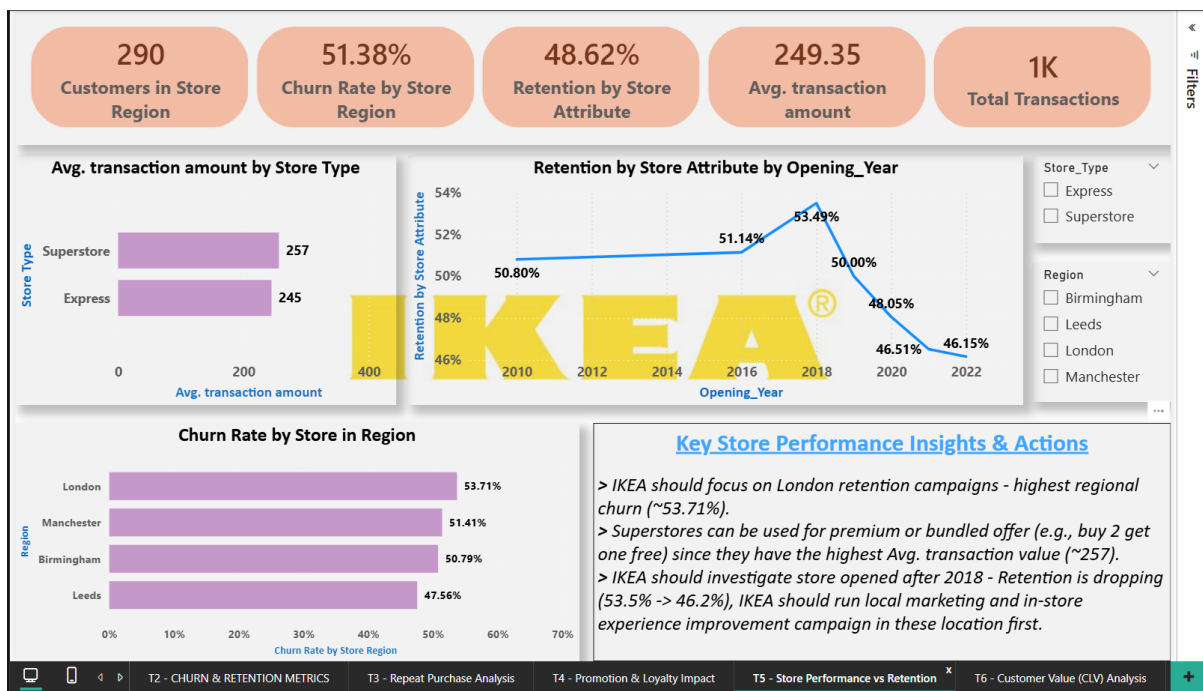
Total Points Earned = SUM (Loyalty_Program[Points_Earned])

Explanation: Sums loyalty points earned across all customers.

Total Points Redeemed = SUM (Loyalty_Program[Points_Redeemed])

Explanation: Sums loyalty points redeemed across all customers.

7. TASK 5 – STORE PERFORMANCE & REGION INSIGHTS



Key findings from store and region analysis:

- Average transaction amount $\approx$ 249.35.
- Superstores have higher average amount ($\sim$ 257) than Express stores ($\sim$ 245).
- Retention is highest for stores opened around 2018.
- Retention drops for stores opened in 2020 and 2022 (around 46%).
- London region has the highest churn ($\sim$ 53.71%).

Business insight:

Superstores are good for premium and bundle offers. London and newer stores need focused retention and customer experience improvements.

Key DAX measures used in Task 5:

Avg Transaction Amount = `AVERAGE ( Customer_Transactions[Amount] )`

Explanation: Average value of a single transaction.

Customers in Store Region = `DISTINCTCOUNT ( Customer_Transactions[Customer_ID] )`

Explanation: Counts distinct customers who have shopped in the selected store region.

Churned Customers in Store Region = `VAR Cust =
VALUES (Customer_Transactions[Customer_ID])`

`RETURN`

`CALCULATE (`
`DISTINCTCOUNT ( Churn_Labelled_Customers[Customer_ID] ),`
`TREATAS ( Cust, Churn_Labelled_Customers[Customer_ID] ),`
`Churn_Labelled_Customers[Churned (Yes/No)] = "Yes"`
`)`

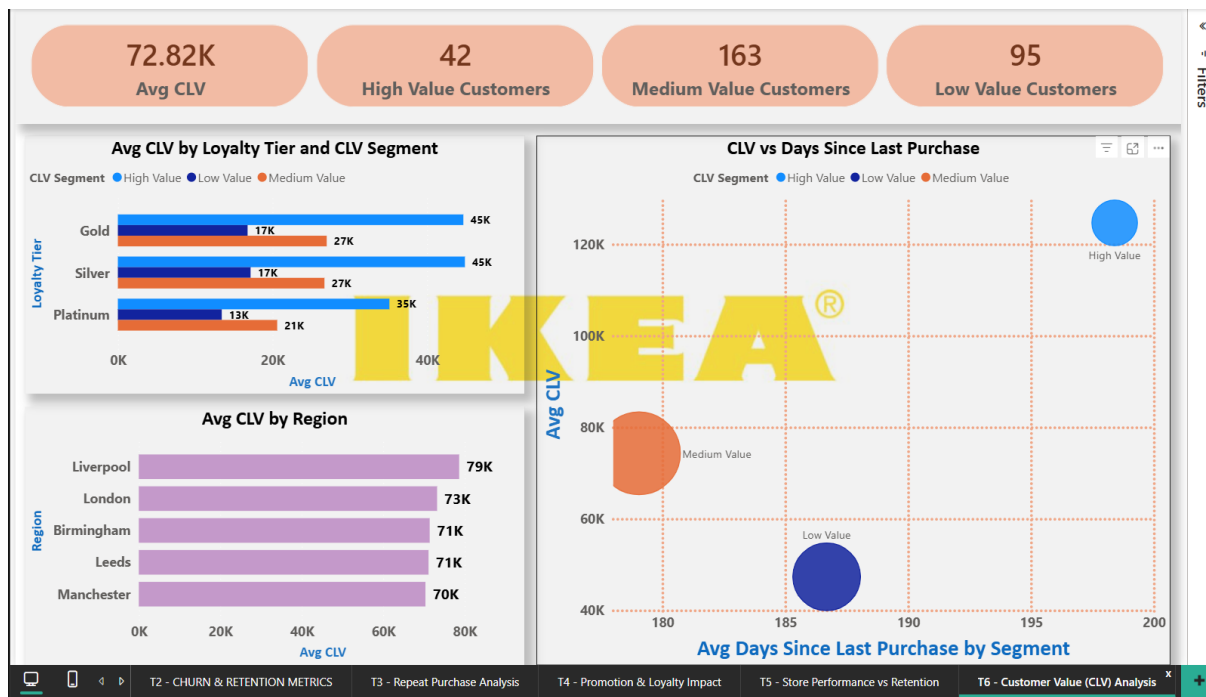
Explanation: Counts churned customers who belong to the selected store region using TREATAS for correct filtering.

Churn Rate by Store Region =

$\text{DIVIDE} ([\text{Churned Customers in Store Region}], [\text{Customers in Store Region}], 0)$

Explanation: Churn percentage based only on customers of that store region.

8. TASK 6 – CUSTOMER VALUE (CLV) ANALYSIS



Key findings from CLV analysis:

- High Value customers = 42.
- Medium Value customers = 163.
- Low Value customers = 95.
- High Value segment has highest CLV but also higher days since last purchase (inactive).
- Silver customers have the highest purchase frequency among tiers.
- Liverpool has the highest average CLV (~79K), Manchester the lowest (~70K).

Business insight:

High-value customers are critical but many are inactive. Silver customers can be grown into higher CLV segments with better engagement.

Key DAX and columns used in Task 6:

Membership Duration (Years) (calculated column) =

$\text{DATEDIFF} (\text{Customer_Demographics}[\text{Membership_Since}], \text{TODAY}(), \text{YEAR})$

Explanation: Calculates how many years each customer has been a member.

Total Amount Spent = SUM (Customer_Transactions[Amount])

Explanation: Total money spent by customers in the current filter context.

CLV_per_customer (calculated column) = VAR TotalAmt =

```
CALCULATE (
    SUM ( Customer_Transactions[Amount] ),
    RELATEDTABLE ( Customer_Transactions )
)
```

VAR DurationYears = Customer_Demographics[Membership Duration (Years)]

RETURN

```
DIVIDE ( TotalAmt, DurationYears, 0 )
```

Explanation: Calculates CLV for each customer as total spend divided by membership duration in years.

Avg CLV =

AVERAGEX (VALUES (Customer_Demographics[Customer_ID]), [CLV_per_customer])

Explanation: Average CLV per customer in the current filter context.

CLV Segment (calculated column) = VAR Rank =

```
RANKX ( ALL ( Customer_Demographics ), [CLV_per_customer], , ASC )
```

VAR TotalCust =

```
CALCULATE ( COUNTROWS ( Customer_Demographics ), ALL ( Customer_Demographics ) )
```

VAR P25 = TotalCust * 0.25

VAR P75 = TotalCust * 0.75

RETURN

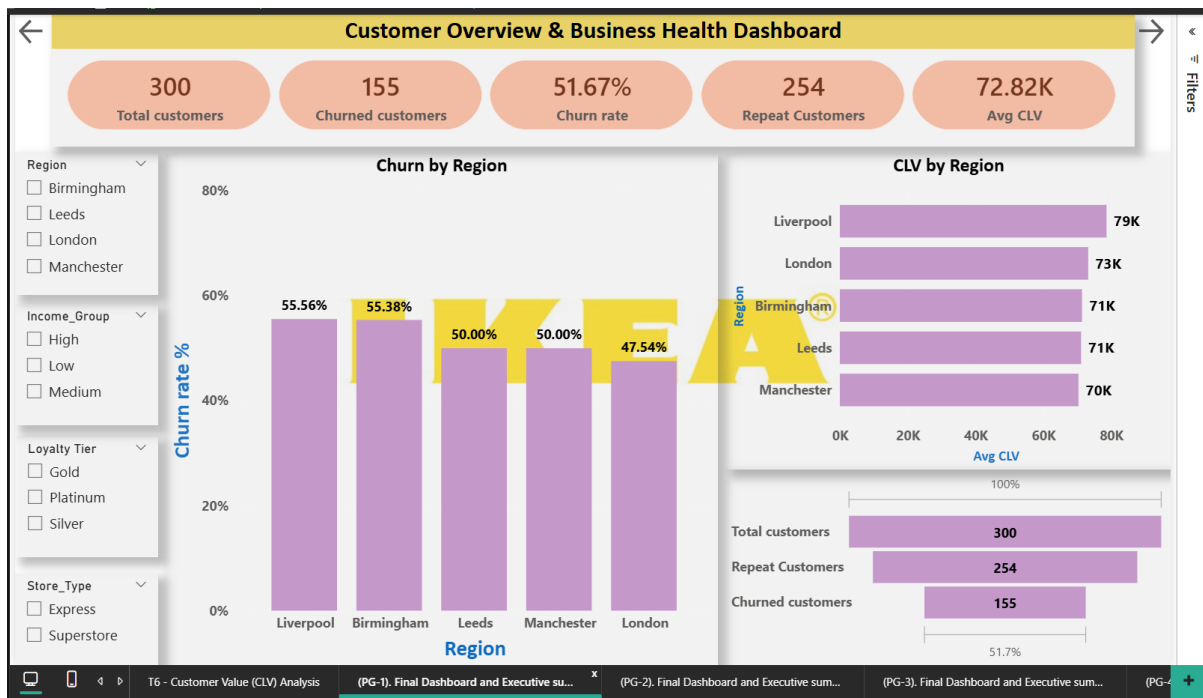
```
IF ( Rank <= P25, "Low Value",
    IF ( Rank <= P75, "Medium Value", "High Value" )
)
```

Explanation: Segments customers into Low, Medium and High CLV based on percentile ranks.

9. TASK 7 – FINAL 4-PAGE EXECUTIVE SUMMARY DASHBOARD

In Task 7, I combined all previous analysis into a final 4-page executive dashboard. This report is designed for business leaders. It summarises customer health, promotion and loyalty impact, store and region performance, and customer value segments, and it highlights where IKEA should focus to improve retention and CLV.

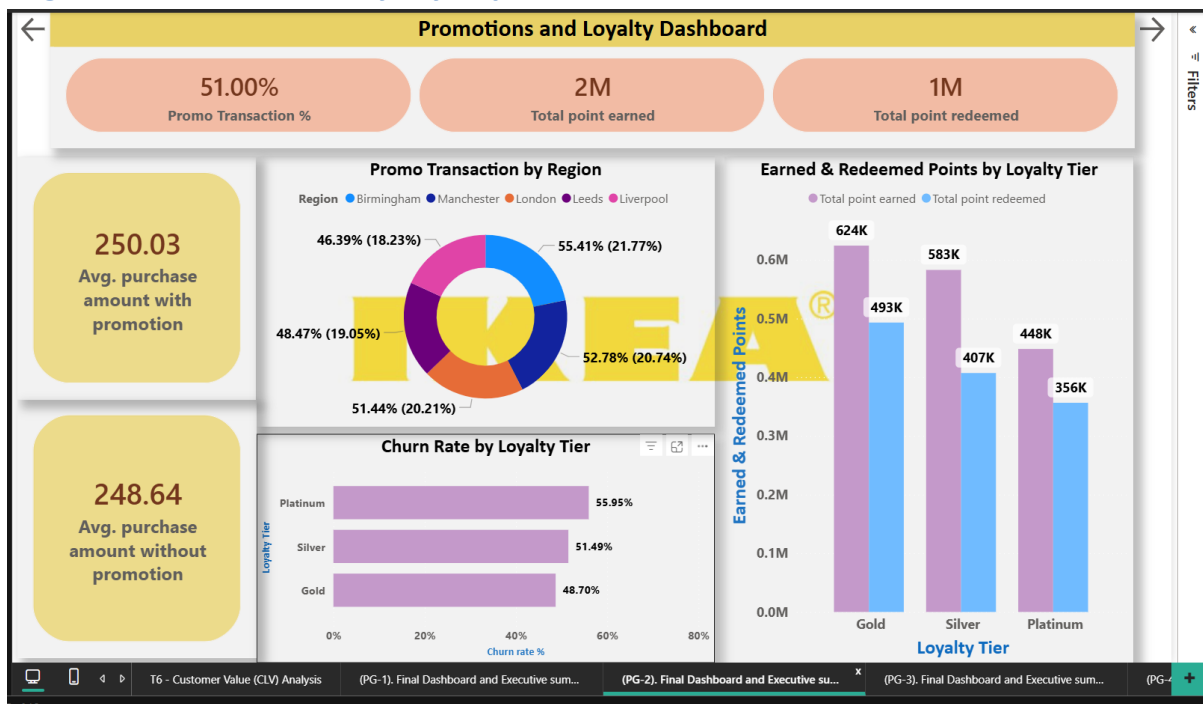
Page 1 – Customer Overview & Business Health



Page 1 gives an overall view of customer health. It shows total customers (300), churned customers (155), churn rate (51.67%), repeat customers (254) and average CLV (72.82K). Charts for churn and CLV by region show that Liverpool and Birmingham have the highest churn while Liverpool also has the highest CLV. A funnel visual shows how many customers remain repeat versus churned.

The main insight from this page is that IKEA has a high churn problem, and the impact is highest in regions like Liverpool and Birmingham where customers are also high value. IKEA should focus on reducing churn in these high-value regions to protect future revenue.

Page 2 – Promotions & Loyalty Impact

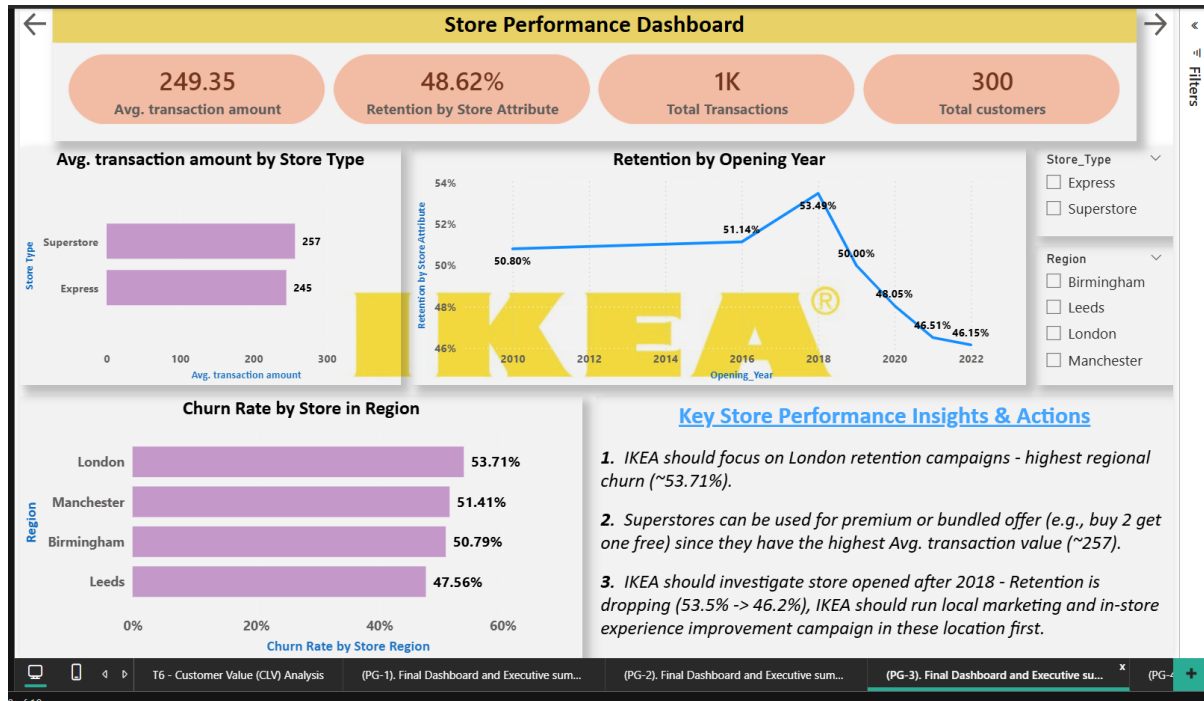


Page 2 focuses on promotions and loyalty. It shows that about 51% of all transactions use a promotion, and the average bill with promotion (250.03) is very close to the bill without promotion (248.64). This means promotions do not reduce customer spending. Charts also

show that Gold and Silver tiers earn and redeem most loyalty points, but Platinum tier still has the highest churn rate, followed by Silver.

The insight from this page is that promotions and loyalty points are being used and are not harming revenue, but loyalty alone is not enough to stop customers from leaving. IKEA needs stronger engagement and retention strategies for Platinum and other high-risk tiers.

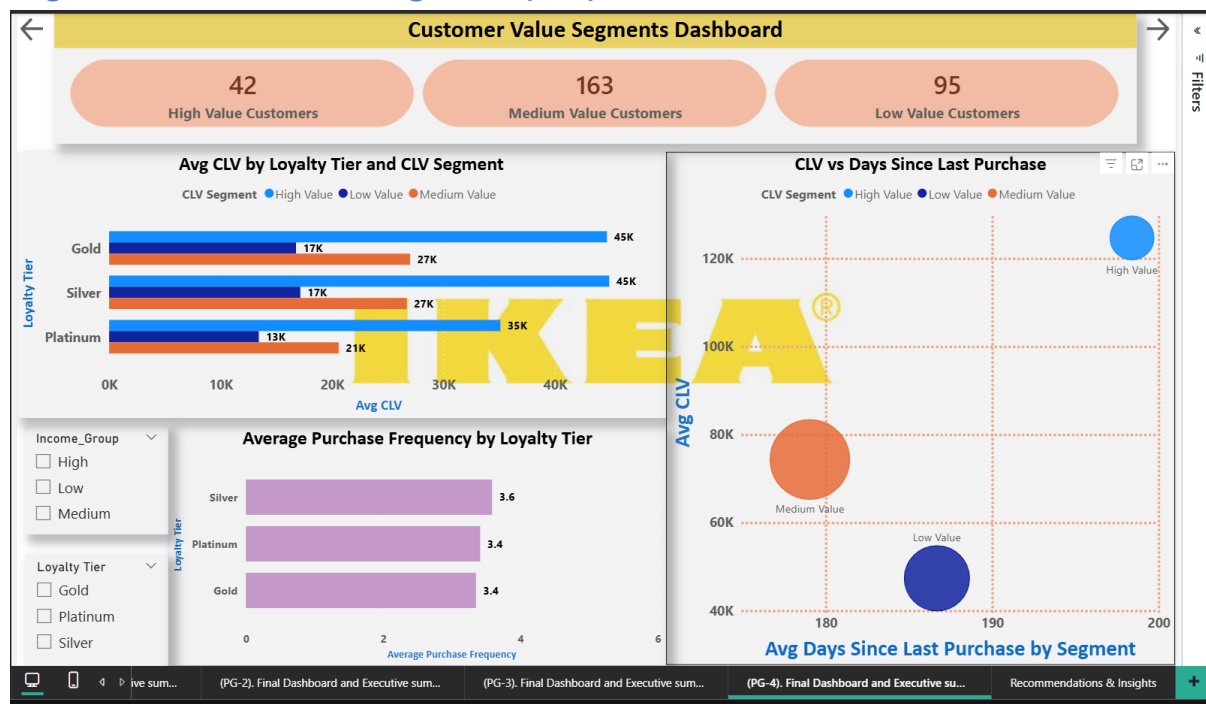
Page 3 – Store & Region Insights



Page 3 looks at performance by store type, opening year and region. Superstores show higher average transaction values than Express stores, which suggests they are better suited for premium and bundle offers. Retention by opening year shows the best performance for stores opened around 2018, while stores opened after 2019 have noticeably lower retention. Churn by region reveals that London has the highest churn rate, followed by Manchester and Birmingham.

The key insight is that London and newer stores opened after 2018 require special attention. IKEA should improve store experience, staff training and local marketing in these locations to raise retention and reduce churn.

Page 4 – Customer Value Segments (CLV)



Page 4 focuses on customer value segments. It shows the count of High, Medium and Low value customers and charts for CLV by loyalty tier and by region. A CLV vs days since last purchase chart summarises how active or inactive each CLV segment is. High-value customers have the highest CLV but also higher average days since last purchase, which means they are valuable but at risk. Silver tier customers have higher purchase frequency but lower CLV, which means they are a growth opportunity.

The insight from this page is that IKEA must protect high-value customers who are becoming inactive and also invest in Silver customers, who are active buyers that can be converted into higher CLV segments through targeted offers and loyalty upgrades.

10. TASK 8 – VIDEO EXPLANATION

Video link:

<https://www.loom.com/share/46a8532e46174891b69267f5608e7b93>

11. KEY INSIGHTS

- Churn is very high (~52%) – more than half of customers are leaving.
- High-value customers are inactive – they have high CLV but have not purchased recently.
- London has the highest churn (~54%) and needs urgent retention work.
- Promotions do not change purchase value much – they are safe but their impact is limited.
- New stores opened after 2018 have poor retention and need better customer experience and marketing.

12. TOP 3 RECOMMENDATIONS

1. Re-engage High-Value but Inactive Customers.
 - They have the highest CLV but higher Days Since Last Purchase.
 - IKEA should offer reactivation rewards, bonus points and personalised offers.

2. Strengthen Loyalty for Silver & Low CLV Groups.

- Early engagement is weak. Silver customers buy often but CLV is low.
- IKEA should improve onboarding and targeted promotions for these groups.

3. Fix High Churn Regions & Underperforming Stores.

- London region has highest churn.
- Stores opened after 2018 show retention drop.
- IKEA should run store-specific improvement and local marketing campaigns in these areas.

13. CONCLUSION

This Power BI project combines customer, transaction, store and loyalty data to analyse churn, repeat purchases, promotions, store performance and CLV. The dashboards help IKEA see which customers, stores and regions need attention and what actions can increase retention and customer lifetime value.